# Matthew Kuney

Kennett Square, PA
484-995-3854 matthew.kuney@gmail.com linkedin.com/in/matthewkuney matthewkuney.github.io

## Objective

To obtain a mechanical engineering internship beginning summer 2027

## Education

**The Pennsylvania State University** **Expected May 2030**
*Bachelor of Science in Mechanical Engineering* *University Park, PA*
GPA: *In progress*

## Relevant Coursework

- Ordinary and Partial Differential Equations
- Matrices

## Experience

**Penn State Learning Factory** **Sep 2026 – Present**
*Learning Factory Teaching Assistant* *University Park, PA*

- Operate and maintain equipment across the wood shop, metal shop, laser engraving, additive manufacturing, and electronics/IoT labs.
- Deliver hands-on safety and equipment trainings, preparing students to independently and safely use Learning Factory machinery.
- Respond to shop emergencies and enforce safety protocols across multiple specialized fabrication labs.

**Mathnasium – The Math Learning Center** **Sep 2025 – May 2026**
*Math Instructor* *Kennett Square, PA*

- Provided one-on-one and small-group mathematics instruction to students across multiple grade levels.
- Helped students develop problem-solving, analytical thinking, and mathematical reasoning skills.
- Communicated with students and staff to track progress and identify areas for improvement.

**Progressive Pool Management Inc.** **May 2024 – Aug 2025**
*Head Lifeguard* *Garnet Valley, PA*

- Managed independent 10-hour shifts, including facility opening and closing procedures.
- Oversaw pool safety, patron services, admissions, and daily facility operations.
- Maintained proper pool chemical levels and performed routine water testing.
- Responded to emergencies and enforced facility safety policies and procedures.

## Projects

**Pick-and-Place Robot** | *Arduino, CAD, Sheet Metal Fabrication, 3D Printing*

- Designed and built a drone-controller-operated pick-and-place robot from scratch, including a welded sheet-metal chassis, drive motors, and a servo-actuated arm.
- Modeled and 3D-printed a custom gripper mechanism to grasp and relocate objects.
- Integrated an Arduino microcontroller with a buck converter and battery system to power and control the robot's motors and electronics.

**Fully Mechanical 3D-Printed Excavator Model** | *Onshape CAD, 3D Printing*

- Designed and fabricated a fully mechanical, non-electronic excavator model in Onshape, driven entirely by springs, gears, nuts, and bolts.
- Engineered a 3-degree-of-freedom arm and bucket assembly, with the cab swiveling on a bevel gear drivetrain.

## Technical Skills & Certifications

**CAD & Design**: Fusion 360, Onshape
**Fabrication**: 3D Printing, Laser Cutting, Metal Fabrication, Woodworking, Welding
**Electronics**: Arduino, Circuit Design
**Other**: Mechanical Design, Prototyping
**Certifications**: Penn State Learning Factory Safety & Equipment Training (2026)

## Leadership / Extracurricular

**ASME (American Society of Mechanical Engineers)** **2026 – Present**
*Member* *University Park, PA*

- Helped organize and run Penn State ASME's Battle Bots competition.